

Software Engineer

Summary

My name is Pravin in India. I am a senior software engineer with a strong focus on Computer Vision, Big Data Analysis, NLP, Flutter and React.js. My work centers on building Computer Vision, scalable, performant, and maintainable applications with modern development practices. My core skill is in Computer Vision and web site development.

I developed many projects which related with computer vision and data analysis part using ML/DL models.

I specialize in creating responsive web applications using React.js and Symfony, leveraging component-based architecture and modern state management techniques. In mobile development, I use Flutter to build high-quality cross-platform applications with smooth performance and consistent UI design.

Through this portfolio, I present projects that demonstrate my approach to Computer Vision, Data processing and to frontend engineering, UI development, and cross-platform application architecture

Skills/Tech Stack

Computer Vision: Yolo, Opencv, OCR, EasyOCR, Pytesseract, Tensorflow, Pytorch and all deep learning model.

Data procesing: Numpy, Pandas, ML, DL and my own model.

Mobile: Flutter, Dart, React-Native, Android, Kotlin

Frontend : React.js, Next.js, Symfony, Javascript/TypeScript, HTML/CSS,

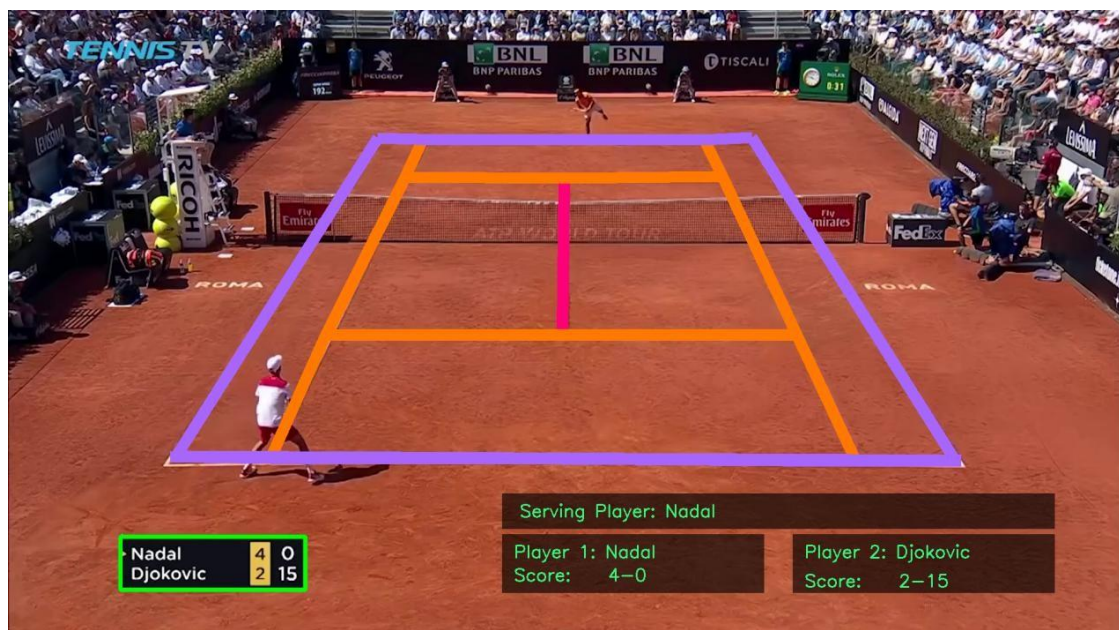
Backend: Node.js, Express.js, Python, FastAPI, Flask, REST APIs, Firebase.

Database: MySQL, MongoDB, SQLite, PostgreSQL

Tools: Git, Docker, CI/CD

Computer Vision && Data Processing.

1. Tennis match analysis project .



In this project, with my expertise skill, I developed tennis analysis project which supports court edge extraction, and player information extraction via scoreboard analysis.

This project involves traditional computer vision skill to detect court edge and OCR engine to recognize the player names, score and finding current serving player.

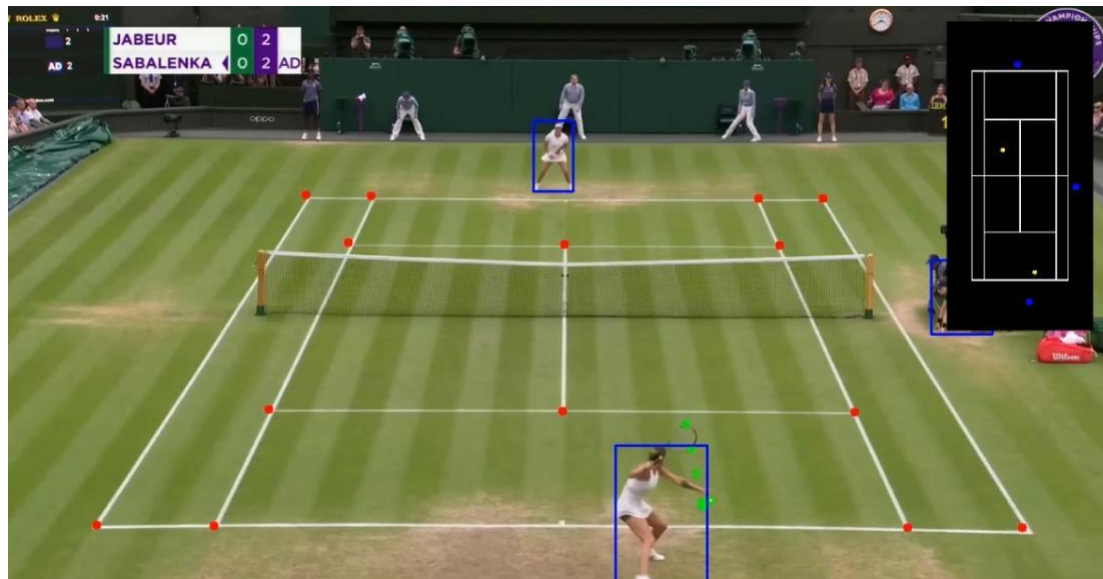
After that I evaluated the average correct match and saved result into csv or google sheet.

2. Tennis ball tracking and speed estimation using Yolo.



In this project, I annotated all frames from video with label using Labelme or Labelstudio.
After that I trained all image to get trained model for detecting ball.
And I detected ball and I added speed estimation part with my own algorithm.

3. Tennis match analysis project.



Tennis match analysis that can process standard tennis match videos and extract valuable insights through computer vision and machine learning techniques.
I developed detection the ball position in consecutive frames, identifies ball bounces, recognize court lines.
Output is coaches and players to analyze patterns, statistics, and overview pattern on right window.

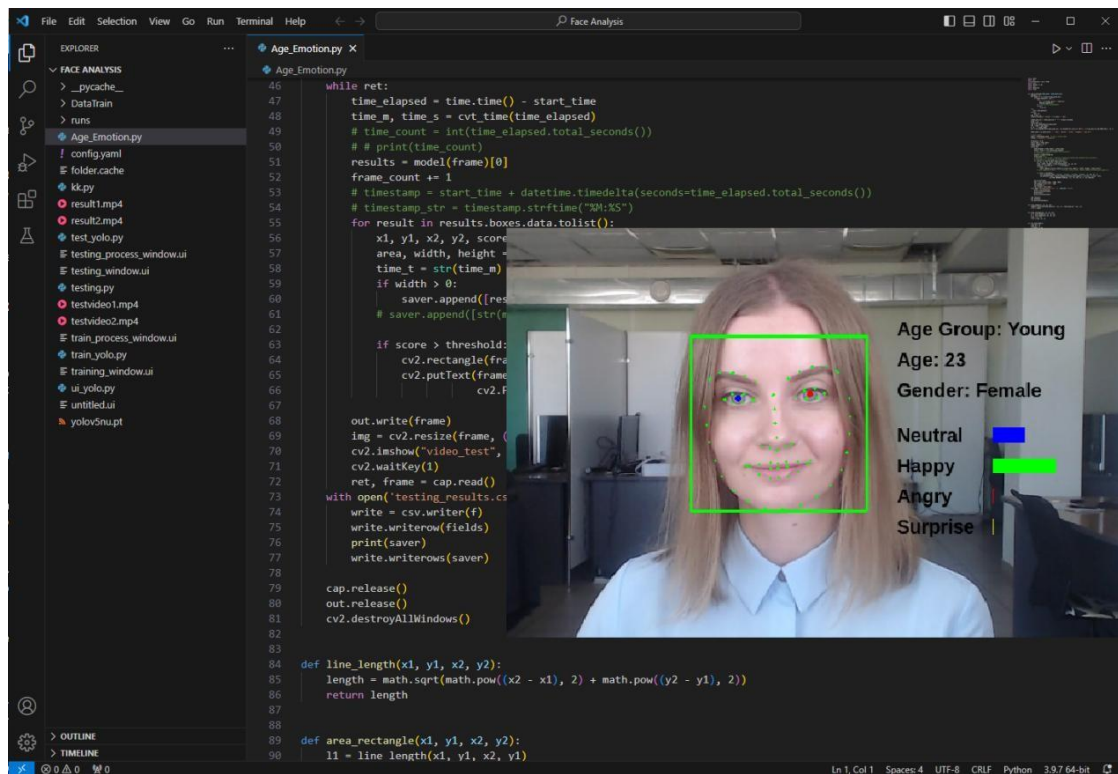
4. Badminton analysis project.



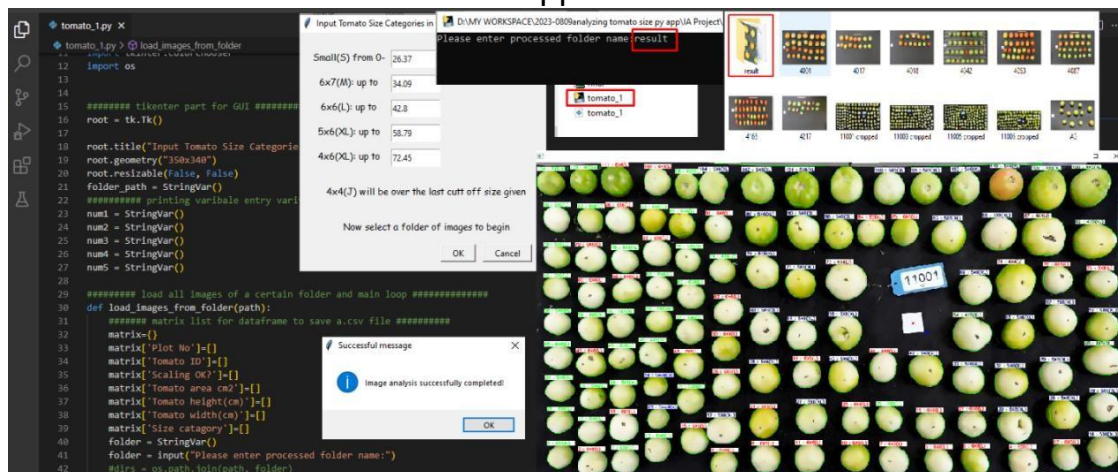
I developed Badminton analysis project which extracts, analyzes and visualizes badminton player movement to generate coach-interpretable tactical insights.
This project analyzes badminton player positioning using trajectory data derived from match footage, pose estimation and shuttle detection.

5. Estimating Age, Gender and Emotions.

This project is to predict exact age number or within an age range, classification male or female and predicts categories such as happy, sad, angry, surprised and so on.
I used CNN to extract facial features and classify, regress them into meaningful output.



6. Tomato size detection windows app.



This project was performed to classify tomatoes via size using traditional image processing. And then count number of each size after that display result with mini GUI.

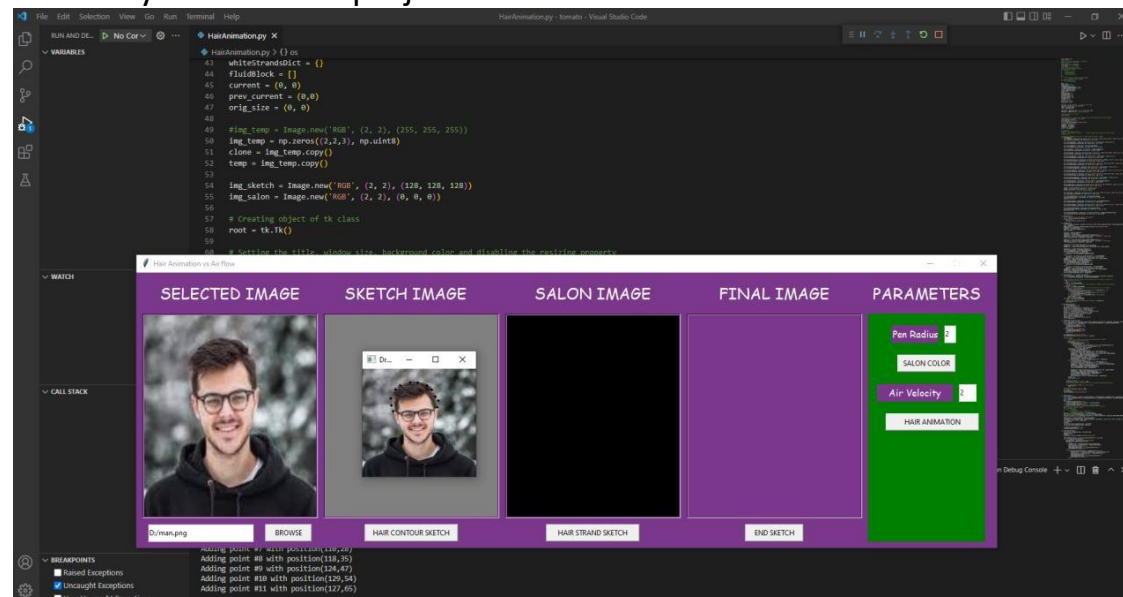
7. Vehicle number plate recognition. (Yolo and OCR)



Firstly, We have to detect Car and then we have to detect number plage region.

We performed OCR engine(EasyOCR or Pytesseract) to extract characters of number plate from cropped image with number plate region. And then we have to display result with screen.

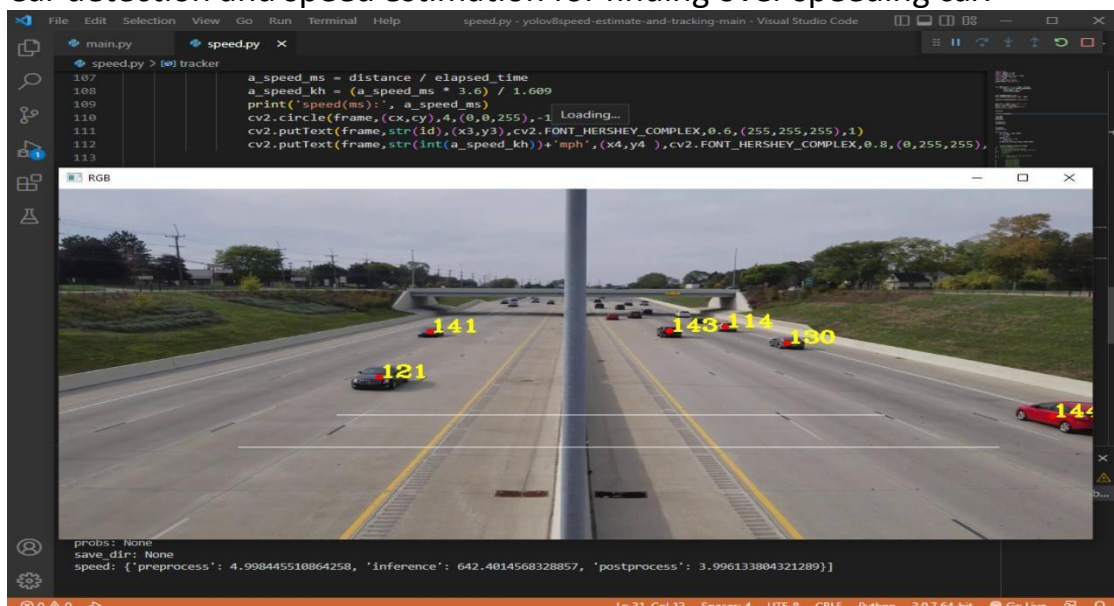
8. Hair style animation project.



In this project, we input original image and image which replaces hair and then we draws mouse on hair of original and run click.

If so, hair is replaced automatically.

9. Car detection and speed estimation for finding over speeding car.

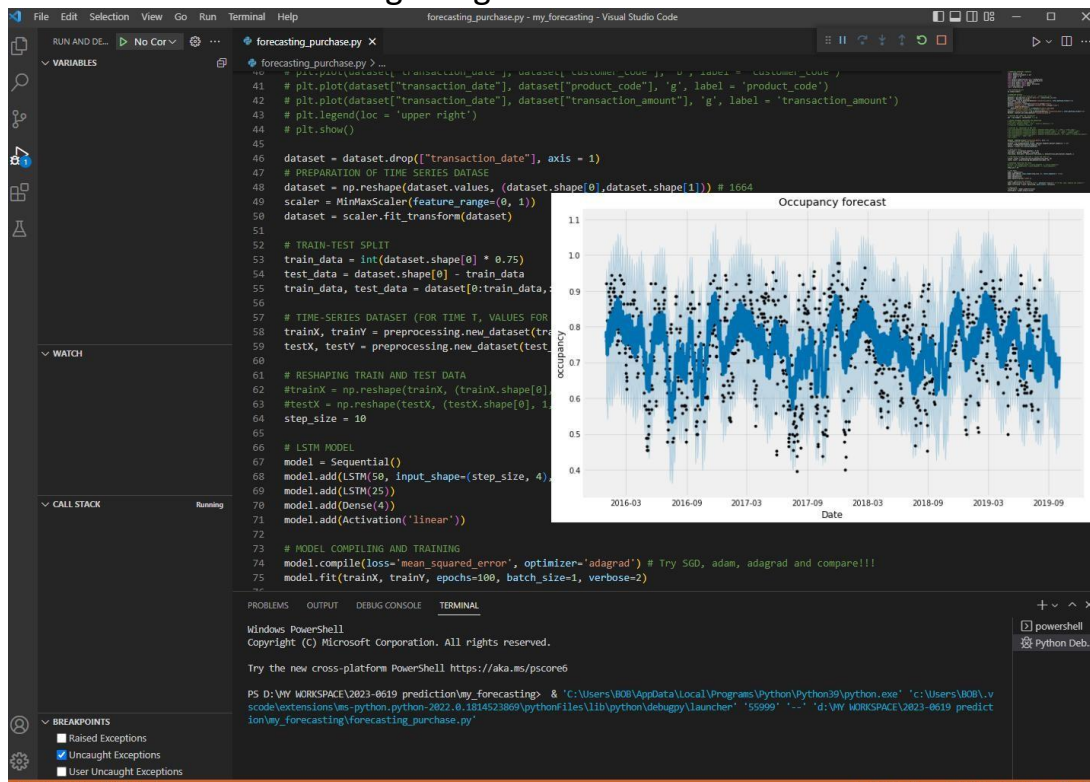


10.

10. Safety stock optimization project (Big data analysis).

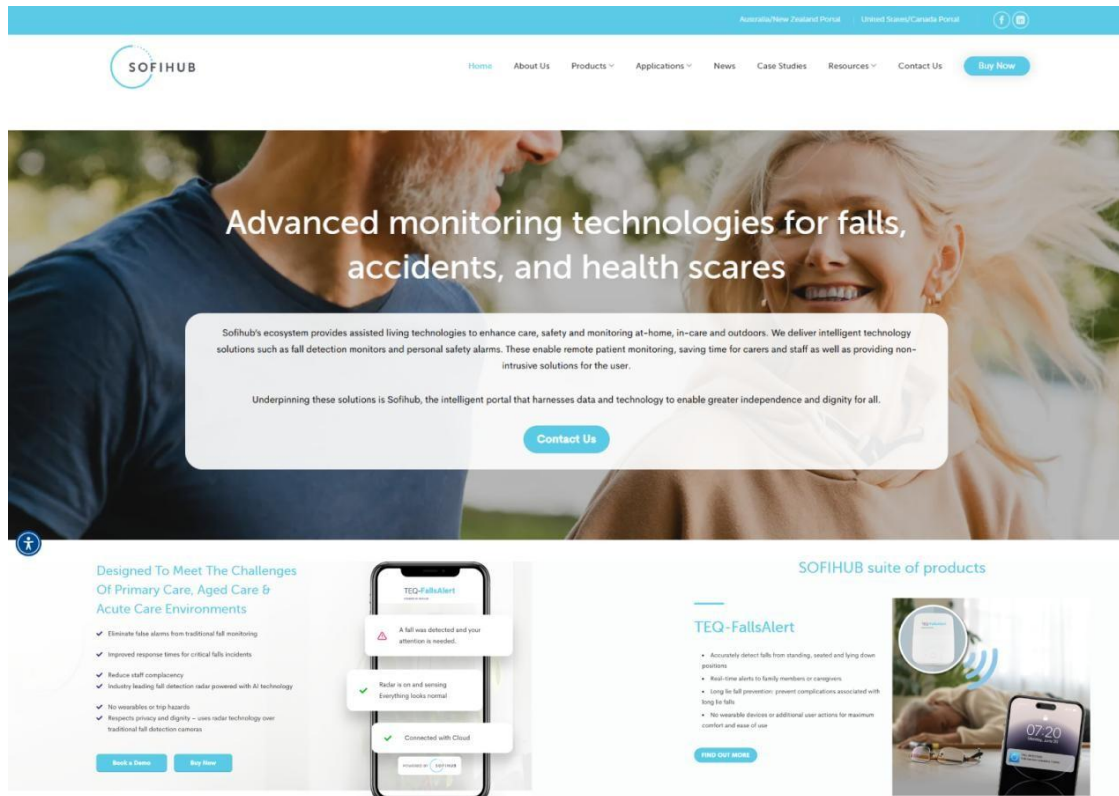


11. Time series forecasting using ML model.

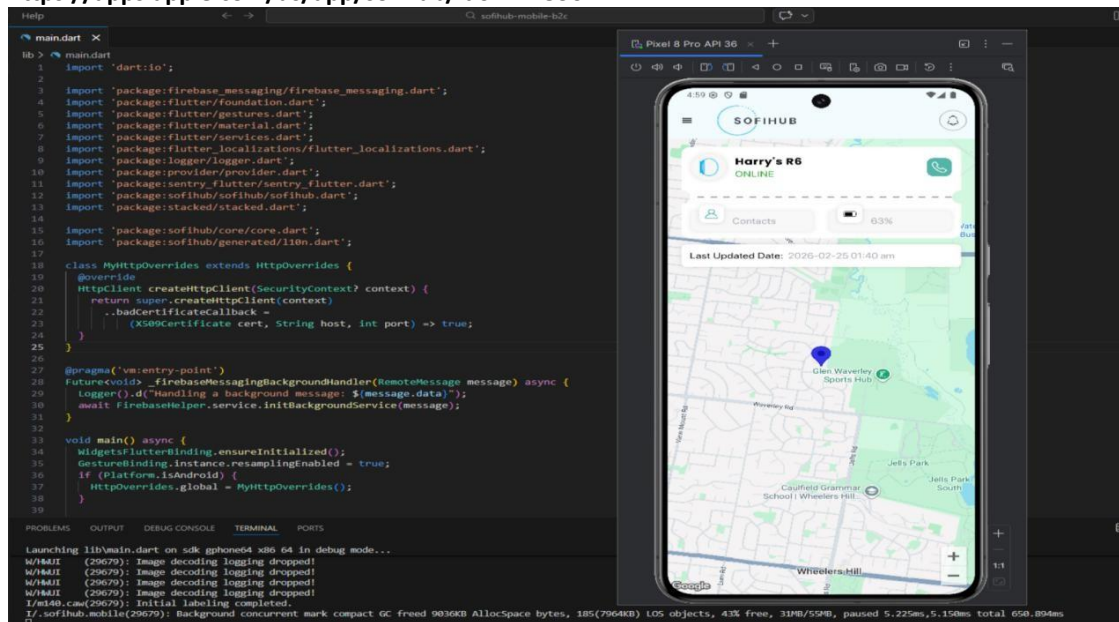


Sofihub | HealthTech Monitoring Platform

Frontend Developer & Mobile Developer



<https://sofihub.com/>
https://play.google.com/store/apps/details?id=com.sofihub.mobile&hl=en_US
<https://apps.apple.com/us/app/sofihub/id6444153627>



Contributed to the development of the Sofihub digital ecosystem, including the official website and cross-platform mobile application used for monitoring safety and health devices.

- Developed responsive and interactive web interfaces for the Sofihub website using modern frontend technologies, ensuring cross-browser compatibility and optimized user experience.
- Built and maintained features for the Sofihub mobile application, enabling users to manage connected safety devices and receive real-time alerts.
- Implemented UI components and application logic for device monitoring, notifications, and device configuration.
- Collaborated with backend engineers and product designers to integrate APIs and deliver a scalable and maintainable application architecture.
- Focused on performance optimization, clean component architecture, and user-friendly design across both web and mobile platforms.

Technologies: Flutter, React.js, JavaScript, REST APIs, Git

SmartShanghai / Smash-Asia – City Guide & Lifestyle Platforms

Frontend Developer & Mobile Developer

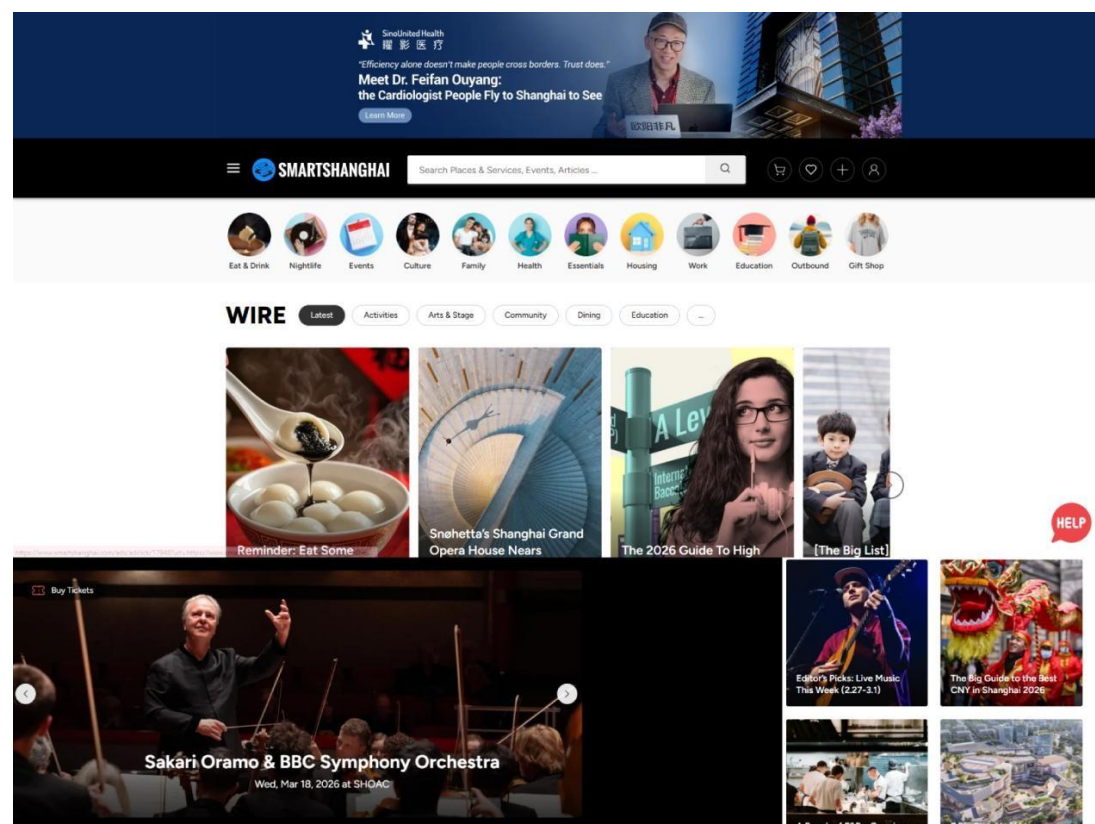
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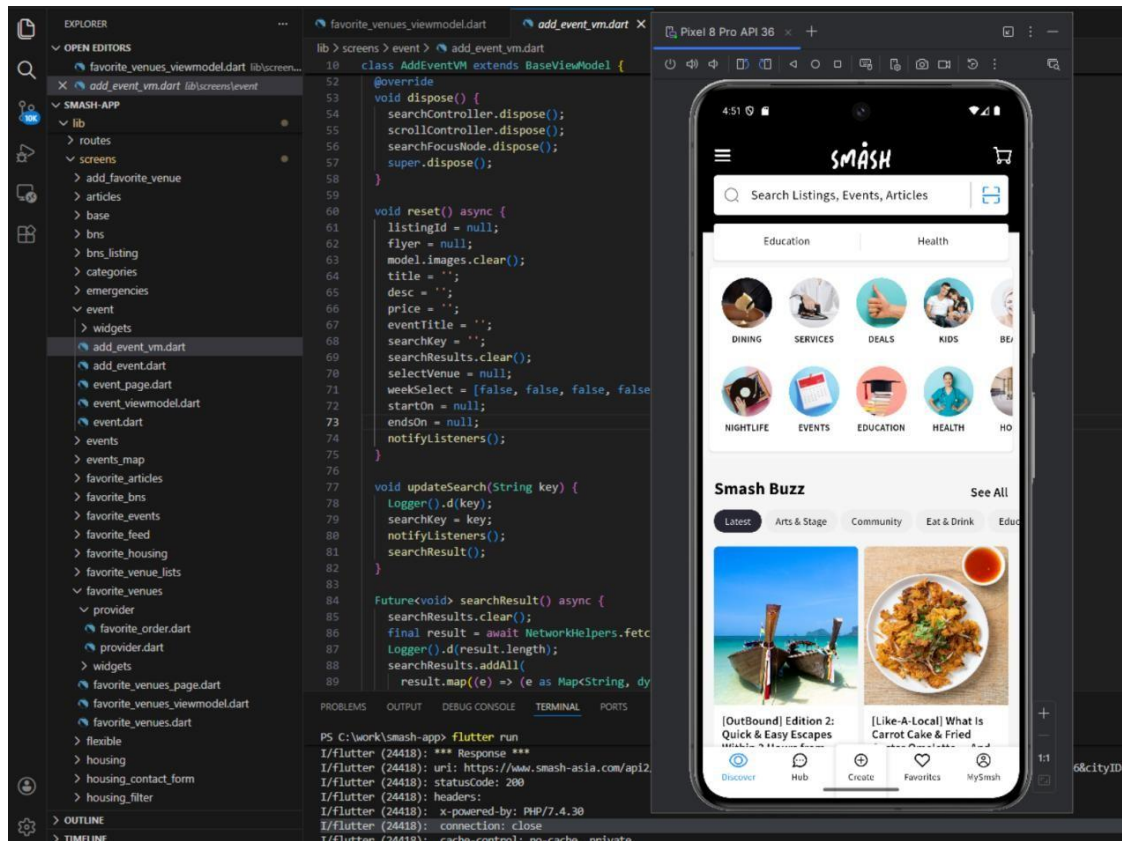
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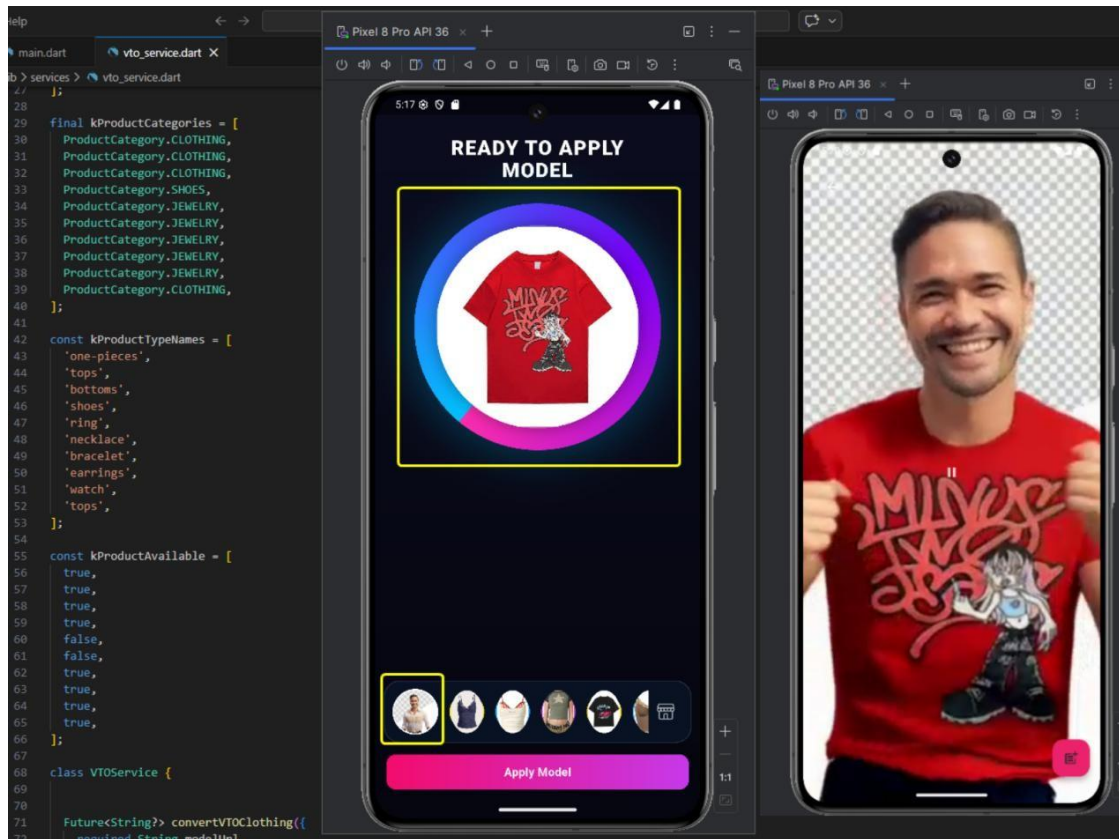
Worked on the development of web and mobile applications for SmartShanghai(for Shanghai) and Smash-Asia(for Singapore), digital platforms that provide city guides, event listings, and lifestyle information for international communities in Asian cities. (The two companies are sister companies.)

- Developed and maintained responsive frontend interfaces for the SmartShanghai and Smash-Asia websites.
- Built and improved cross-platform mobile app features for iOS and Android applications used to explore venues, events, and local services.
- Implemented UI components for venue directories, event listings, search functionality, and location-based discovery features.
- Integrated frontend and mobile interfaces with backend APIs to deliver dynamic content and real-time data updates.
- Collaborated with designers and backend engineers to ensure consistent user experience across web and mobile platforms.

Technologies: Flutter, Symfony, React.js, JavaScript, REST APIs, Git

Mobile Developer & AI Engineer

(Not yet released)



Developed mobile applications integrating AI-powered Virtual Try-On (VTO) technology and cryptocurrency wallet functionality using Flutter.

- Designed and developed cross-platform mobile applications using Flutter for both Android and iOS platforms.
- Built and integrated AI-based Virtual Try-On (VTO) features, enabling users to visualize products through machine learning models.
- Created and deployed AI models using HuggingFace for image processing and inference within the mobile application.
- Integrated AI inference pipelines with Flutter applications to deliver real-time Virtual Try-On experiences.
- Implemented core cryptocurrency wallet features including wallet creation, transaction management, and secure asset handling.
- Worked with blockchain APIs and secure cryptographic libraries to support digital asset transactions and wallet management.
- Collaborated with backend and product teams to ensure seamless API integration and scalable application architecture.

Technologies: Flutter, Dart, HuggingFace, Machine Learning, Blockchain APIs, REST APIs, Git

And other sites are below...

Product selling website using WordPress.



Product selling website

